

Leadership Succession and Militant Extra-Lethal Violence

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Abstract

Why do armed groups use extreme forms of violence such as beheadings, despite their significant costs? This study argues that leadership transitions create authority crises that incentivize successors to adopt extra-lethal violence as a tool of internal consolidation and external signaling. These pressures are particularly acute for successors with prior leadership experience in other armed groups: having previously lost power, these leaders face additional reputational deficits and are especially likely to view extra-lethal violence as a useful instrument of authority consolidation. Drawing on an original dataset of 206 leaders from 108 jihadist groups active between 1976 and 2023, we find that organizations are significantly more likely to use beheadings under successor leaders than under founders. This effect is most pronounced among those with prior rebel leadership experience. We also find that these patterns are more consistent with successors' strategic use of beheadings to address short-term authority deficits than with alternative explanations such as ideological extremism, technical skill, transnational network ties, or unsanctioned violence by subordinates. By shifting attention from organizational incentives to leader-level dynamics, this study contributes to research on militant leadership, succession in armed groups, and the strategic logic of extra-lethal violence.

Keywords: militant leaders, succession, leader decapitation, jihadist organizations, beheading

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Introduction

Periods of leadership transition are fraught with uncertainty for clandestine organizations. Whether the group is an ethno-separatist insurgency, a drug cartel, or a transnational jihadist movement, the death, arrest, or retirement of a founding figure often generates challenges for the successor who steps into the void. Lacking the charisma or historical legitimacy attached to the founder, successors must convince internal rivals of their authority and external audiences of their organization's continued vitality. How, then, do successors tackle the challenges of building reputations and maintaining organizational control after assuming power?

This study argues that the a group's repertoire of violence, particularly extra-lethal violence, becomes a strategic instrument in this struggle. Take, for example, Jemaah Islamiyah (JI). Formed in the 1990s to violently overthrow the Indonesian regime and establish an Islamic State, JI gained prominence under the leadership of its founder, Abdullah Sungkar. Following Sungkar's death in 1999, the group, now under new leadership, escalated its violent campaign (Jones 2005). This escalation included the use of a new and more gruesome form of violence. In 2005, JI carried out a high-profile beheading attack against three Christian schoolgirls in 2005 (Hwang 2019).

Extra-lethal violence occurs in conflict settings around the world. Jihadist organizations in multiple regions, leftist and separatist rebel groups in Nepal, Thailand, and Vietnam (Acharya and Muldoon 2017; Berni 2023), criminal organizations in Mexico (Phillips 2018), and violent right-wing actors in Russia (Koch 2018) have used beheadings and other forms of extra-lethal violence in recent decades. Existing research argues that armed groups¹ use extreme forms of violence instrumentally—to generate publicity (Pape and Morrell 2014), provoke foreign states (Zech and Kelly 2015), deter certain forms of behavior (Tinnes 2016; Brzezinski 2024), or intimidate civilians (Campbell 2014). While shedding critical light on militant behavior, this scholarship often treats armed groups as unitary actors, focusing solely on organizational incentives to employ extreme violence.

We argue that militant leaders—individuals who occupy the top leadership position and exert ultimate decision-making authority over organizational policies of armed groups (Prorok 2016)—may also play a role in shaping groups' use of beheadings. Leadership transitions often plunge armed groups into periods of instability, internal conflict, and external pressure, creating multiple challenges for successor leaders (Bacon and Grimm 2022; Mendelsohn 2024). We argue that, under pressure to assert internal control and project strength externally, successors may turn to extra-lethal violence to bolster their internal authority and signal the group's continued cohesion and capacity to external audiences. Yet, this dynamic is particularly acute for successors who

¹We define armed groups as “violent non-state groups that challenge their host-state to achieve a political objective” (Loyle et al. 2023, 265)

previously led other armed groups. These leaders not only draw on past experiences managing rebellion, but also carry the burdens of earlier failure—having been ousted, arrested, or defeated in their previous group. Consequently, they may be more inclined to see extra-lethal violence as a strategic and necessary instrument to signal resolve.

We test our argument on the use of beheadings by jihadist organizations. We do so for multiple reasons. First, analyzing a subset of armed groups that share similar ideological commitments allows us to come closer to isolating the effects of leadership dynamics from ideological variation. Second, jihadist groups have been frequent targets of state-led leadership decapitation campaigns (Price 2012; Jordan 2014), providing opportunities for examining how new jihadist leaders navigate the challenges of succession. Within the broader category of extra-lethal violence, we concentrate on beheadings due to their disproportionate human toll: in the past two decades alone, jihadist groups have reportedly beheaded nearly 5,000 individuals (Brzezinski 2024). This prevalence, coupled with the unique spectacle of beheadings, makes them an important form of violence to understand.

Focusing on the internal dynamics of jihadist groups, we draw on original data on the backgrounds of 206 top leaders of 108 jihadist organizations active between 1976 and 2023 across 33 countries in Central Asia, the Middle East and North Africa, South and Southeast Asia, and sub-Saharan Africa (Amjad et al. 2025). We combine this leader-level data with recent event-level data on beheadings perpetrated by jihadist groups to create a leader-year-level dataset of over 1,300 observations (Brzezinski 2024; Berlin 2024). Our analysis yields two key findings. First, jihadist organizations are significantly more likely to engage in beheadings under successor leaders than under founding leaders. Second, this effect is particularly pronounced among successors with prior leadership experience in another armed group. Additional analyses probing the causal mechanisms suggest that these findings are unlikely to be driven by alternative explanations—such as a leader’s ideological extremism, technical skill, transnational network ties, or unsanctioned violence by insubordinate fighters—and instead reflect successors’ strategic use of beheadings to address short-term authority deficits following leadership transitions.

This article makes at least three contributions. First, our focus on militant leaders builds on the burgeoning strand of scholarship that highlights the importance of individuals in conflict processes (Doctor 2020; Bacon and Grimm 2022; Huang, Silverman and Acosta 2022; Ortiz 2024). Despite the central role of leaders in guiding tactics and strategy, first-image explanations of armed group behavior remain relatively rare (Bacon and Arsenault 2019), as conflict scholarship “largely sustains a blind spot regarding how differences in rebel leaders’ individual characteristics affect [conflict] outcomes” (Acosta, Huang and Silverman 2023, 2). Pushing forward this emerging agenda, we theorize why individual leaders have incentives to sanction costly forms of violence to consolidate their hold on power.

Second, this is one of the first studies to systematically examine variation in armed group behavior under founding leaders and their successors. Recent scholarship has analyzed the distinction between founding leaders and successors in armed groups (Bacon and Arsenault 2019; Mendelsohn 2022, 2024). However, quantitative work on this question remains rare, with few exceptions (Tiernay 2015; Prorok 2016; Blair, Horowitz and Potter 2022). Focusing on succession dynamics within armed groups builds on research exploring how previous military training (Doctor, Hunter and Ligon 2024), education (Acosta, Huang and Silverman 2023), and age (Silverman, Acosta and Huang 2024) may affect militant leaders' decision-making.

Third, we contribute to scholarship on extra-lethal violence (Fujii 2013; Holmes 2022; Badillo-Sarmiento and Trejos-Rosero 2025). Militant groups have decapitated thousands of people in numerous conflict zones, and the brutal nature of beheadings influences public opinion and states' counterterrorism strategies (Friis 2015). A 2014 Wall Street Journal/NBC News poll found that more U.S. citizens were aware (94 percent) of the Islamic State's (IS) beheading of James Foley than any other major event in the previous five years (McCalmont 2014). The following section explores the dynamics surrounding why and when armed groups may use extra-lethal forms of violence like beheadings.

Why Armed Groups Use Extra-Lethal Violence

Extra-lethal violence denotes acts of face-to-face violence that transgress shared norms surrounding the proper treatment of the living and the dead (Fujii 2013). This broad category includes violent acts such as crucifixion, immolation, maiming, and beheading and may be perpetrated by both state and non-state actors (Fujii 2013). Such violent acts often generate significant attention and may be publicly staged for that purpose (Fujii 2021). After all, armed groups have publicly beheaded or filmed beheadings in disparate contexts. In 2020, for example, armed actors in Mozambique beheaded more than 50 people, turning a village football pitch into an "execution ground" (BBC 2020).

The use of extra-lethal violence carries significant costs for armed groups. For one, this violence can cost armed groups local support. The beheading of seven Hazara civilians by Islamic State Khorasan in Afghanistan in 2015 reportedly sparked "a furious wave of reaction on social media" and widespread protests (Andalid 2015). IS media content that featured brutal violence also decreased public support for the group (Mitts, Phillips and Walter 2022). Armed groups themselves have recognized how extreme violence can erode popular support. In an internal al-Qaida letter, Ayman al-Zawahiri stated: "Among the things which the feelings of the Muslim populace who love and support you and will never find palatable...are the scenes of slaughtering the hostages..." (Al-Zawahiri 2005).

Using extra-lethal violence may also increase counterterrorism pressure on armed groups. For example, Egypt launched airstrikes in 2015 against the Islamic State in Libya after the group beheaded 21 Egyptians, with Egyptian President Abdel Fattah al-Sisi declaring: “Egypt preserves the right to respond, in the appropriate manner and timing, *to exact retribution*” (Kirkpatrick 2015). As Fujii summarizes (2013, 410), the central puzzle surrounding such violence is why it occurs at all: “If the goal is to terrorize or subdue, why terrorize through sexual torture or burning victims alive...when there are alternative terrorizing acts — sniper attacks, reprisal killings, mass shootings, disappearances — that do not transgress norms so brazenly” and require less time and effort?

Despite these costs, extra-lethal violence remains a recurring feature of contemporary conflict. In addition to leftist, separatist, and right-wing actors, nearly 40 percent of jihadist groups have employed beheadings at some point in their histories, with these incidents occurring across a wide range of conflicts and over sustained periods. However, the specific conditions under which armed groups choose to use extra-lethal violence remain underspecified. Although few theories speak directly to beheadings, scholars have outlined broader strategic logics to explain why armed groups might engage in costly violence. The following section reviews these explanations, outlining why groups may view extra-lethal violence as a strategically useful tool. These accounts—rooted in organizational-level incentives—provide a basis for our argument, which shifts attention to leader-level factors that shape whether and when groups turn to extreme violence.

Existing Explanations

Outbidding remains “one of the most widely cited theories of terrorist groups’ motivations and actions” (Farrell 2020, 7). The theory of outbidding posits that competition between armed groups can lead to violent escalation. In pursuit of achieving a “market share” of support, organizations may increase the scale and shock value of their violence to demonstrate resolve (Bloom 2004). In addition to intensifying attacks, groups might also diversify their violence to differentiate themselves from rivals (Horowitz, Perkosi and Potter 2018). Although most of the work on outbidding focuses on other tactics, armed groups may similarly turn to beheadings when facing heightened competition to signal greater resolve to potential supporters.

Similarly, groups may use extreme violence to generate publicity and attract recruits (Pape and Morrell 2014). Beheadings can generate significant publicity: “The media attention that comes with a publicized execution is likely to help an organization in the long term to generate the publicity required to recruit, compete with rivals, and raise money from sympathetic donors” (Loertscher and Milton 2018, 5). Smaller or struggling groups may be more likely to behead to increase visibility to potential recruits.

Beheadings may also serve as tools of intimidation. Groups may employ extra-lethal violence

to deter disobedience and intimidate civilians (Badillo-Sarmiento and Trejos-Rosero 2025; Campbell 2014). As Holmes (2022, 370) writes: “The effects of such violence go beyond simple lethality and serve as an extravagant display of power” (Holmes 2022, 370). Several jihadist beheadings have aimed to punish or deter civilian collaboration. In 2008, for example, the Pakistani Taliban publicly beheaded two Afghans accused of spying, warning that “whoever spies for the Americans will meet the same fate” (Bahauddin 2008). The use of beheadings as intimidation may also correlate with territorial control (Brzezinski 2024).

In addition to intimidation, beheadings may be used to provoke an adversary into a disproportionate military response. Groups may seek disproportionate retaliation to radicalize moderates and mobilize support by demonstrating that an enemy is untrustworthy (Kydd and Walter 2006, 69). Zech and Kelly (2015, 88), for example, describe IS’s beheadings as “a tactic intended to provoke repression by Western powers.” Beheading foreign citizens, for instance, may thus serve as a deliberate provocation designed to expose perceived hypocrisy or brutality among foreign state actors.

Internal organizational dynamics offer further explanations. Leadership decapitation or other shocks may reduce leaders’ control over rank-and-file behavior, increasing the likelihood of extreme violence (Abrahms and Potter 2015). Similarly, groups with economic endowments—such as access to lootable resources or state sponsorship—are often less reliant on civilian support and more likely to attract opportunistic recruits with fewer incentives to restrain violence (Weinstein 2006). If principal-agent problems drive brutality, however, we might expect leaders to publicly disavow unauthorized beheadings carried out by lower-level members.

Group ideology may also play a role in behavior and tactical choices. Ideology denotes a “more or less systematic set of ideas that identifies a referent group, voices the grievances that this group faces, articulates objectives, and delineates a program of action” (Gutiérrez-Sanín and Wood 2017). Ideology can impact how and why armed groups employ particular forms of violence (Revkin and Wood 2021). Groups with ideologies that define enemies in religious or moral terms—e.g., “apostates” or “imperialists”—such as those espoused by transnational jihadist movements, may be more likely to employ beheadings as a form of symbolic retribution.

Finally, ties to external organizations can shape tactical repertoires. Among jihadist organizations, beheadings have become “a signature element and key feature of the IS ‘brand’ of terrorism, distinguishing the group from other jihadist and secular terrorist actors throughout the world” (Tinnes 2016, 78). Organizations that pledge allegiance to IS may adopt similar methods to signal alignment and increase the likelihood of being formally recognized as affiliates (Brzezinski 2024).

While providing critical insight into patterns of armed group violence, existing theories have notable limitations in explaining why and when groups are more likely to employ beheadings. First, many arguments struggle to account for the timing of adoption. An organization’s ideology,

levels of resources, or number of rivals can remain relatively constant even as the group begins to use beheadings. For instance, numerous jihadist groups have avoided the practice despite shared ideological foundations (Brzezinski 2024) and its permissibility under certain conditions in Islamic jurisprudence (Euben 2017, 1027-1028).

Other prominent theories also fall short in explaining beheading dynamics. While civilian victimization has been linked to levels of territorial control, groups such as Jund al-Khilafa in Algeria began beheading captives before controlling territory. Publicity- and provocation-based explanations are similarly limited: only a small percentage of jihadist beheading victims in recent years have been citizens of the U.S. or Europe (Berlin 2024). Likewise, competition-based theories are undermined by cases where groups delayed beheading long after rivalries emerged. For instance, the Islamic Movement of Uzbekistan began beheading years after facing competition, while the Moro Islamic Liberation Front disavowed responsibility for a 2007 beheading incident despite competing for local support (Dacanay 2007). And while IS popularized beheadings, several groups employed the tactic prior to its rise, while others who pledged allegiance—such as Ansarul Khalifa Philippines and Jamaah Ansharut Daulah—have never used it.

More broadly, existing theories focus primarily on organizational-level explanations, treating armed groups as unitary actors. This perspective overlooks variation in the preferences and incentives of different types of militant leaders to engage in particular tactics. Addressing the limits of current accounts opens space for new theory-building around leadership and the strategic use of extra-lethal violence.

Successor Leaders Face Authority Crises

Militant leaders exert profound influence on the behavior of armed groups. The clandestine nature of these organizations allows leaders to “wield enormous power and influence over all aspects of their organizations, from their structure and identity to the pace and scale of group activities” (Price 2012, 16). Leaders shape organizational trajectories by articulating ideological visions, defining long-term objectives, crafting strategies and tactics, and managing relationships with external actors (Freeman 2014). Among these leaders, founders often play an especially important role in shaping the strategic direction of armed groups.

Founders—the first top leaders of armed groups—shape organizational tactics, resource mobilization strategies, and discursive framing efforts (Bacon and Grimm 2022). They typically establish their reputations during wartime, cultivating trust, credibility, and centrality to the organization’s internal order over the course of the rebellion (Meng and Paine 2022). Founders may also command authority through charisma, inspiring members to follow and sacrifice for their vision (Freeman 2014; Mendelsohn 2022). Within the Shining Path, Abimael Guzman possessed

“a deity-like image among many group members” (Mendelsohn 2024, 16). Mullah Omar similarly developed “quasi-mythical image” during his tenure as the Taliban’s leader (Ruttig 2021). James D. Ellison, founder of the Covenant, the Sword, and the Arm of the Lord, was regarded by members of the far-right group as divinely chosen to lead (Hofmann 2015).

The charisma or historical legitimacy of founders can endure long after their departure from power. Research on business leadership, for instance, shows that founders leave a strategic legacy on their companies after being replaced (Harris and Ogbonna 1999). Similarly, founders of armed groups can retain their “quasi-mythical” status even after their death. Osama bin Laden, for example, remained one of the most frequently cited figures in al-Qaida propaganda years after his death (Aboud et al. 2023). In contrast, successor leaders, who lack comparable storied legacies or personal connections to groups’ fighters, may have to pursue alternative strategies to preserve cohesion and inspire obedience.

Given the prominence of founding leaders, leadership succession marks a period of uncertainty and vulnerability (Price 2012; Johnston 2012; Ryckman 2020; Blair, Horowitz and Potter 2022). Leadership transitions can disrupt internal order, exacerbating instability in the aftermath of a command change (Jordan 2014). Unlike institutionalized settings with routinized succession procedures, armed organizations frequently lack designated successors or codified mechanisms for leadership transfer, such as elections (Cunningham and Sawyer 2019), making each transition a potentially contested event in intra-group politics.

Successor militant leaders face numerous challenges upon assuming command. Beyond inheriting authority under unstable conditions, they often face reputational deficits, doubts about their resolve, and the risk of internal dissent or punishment (Bacon and Grimm 2022; Mendelsohn 2024). New leaders often lack established reputations, or “the subjective beliefs that actors use to predict future behavior” (Lupton 2018). This absence of a behavioral track record makes it difficult to discern a leader’s “type” upon their ascension to power (Horowitz and Fuhrmann 2018). At the same time, recent research suggests that leaders are often “concerned,” and “even obsessed,” with their statuses and reputations (Dafoe, Renshon and Huth 2014). Moreover, reputations for resolve are more likely to adhere to individual leaders—rather than to organizations—when those leaders are perceived as central to policy formation and execution (Renshon, Dafoe and Huth 2018).

In the face of such uncertainty, successors are also vulnerable to internal contestation and external sanction. Militant leaders are subject to evaluation by organizational elites and may face resistance from rivals, especially during periods of strategic ambiguity following leadership transitions (Rigterink 2021, 34). Senior figures—whose status was consolidated under the founder—may fear displacement or diminished influence under the new leadership. As Prorok (2016) notes, militant leaders must retain the support of internal elites, who can withdraw loyalty or punish leaders².

²Even in decentralized groups, leaders remain vulnerable to internal challenges like coups or shifts in power toward

Similarly, mid-level commanders and frontline fighters may retain an ideological commitment or personal loyalty to the founder (Downton 1973), leading them to view the successor with suspicion. Thus, successor leaders face the classic principal-agent problem: inheriting an organization composed of operatives whose preferences or loyalties may diverge from their own (Abrahms and Potter 2015).

Without the legitimacy derived from founding the group or defining its mission, successors may struggle to establish and maintain authority (Rigterink 2021). For instance, many Taliban members opposed the leadership of Mullah Mansoor following his succession of Mullah Omar: “Not only did Mansoor initially fail to get the support of the family of Mullah Omar, he was also criticized by senior Taliban leaders” and military commanders (Hamming and Roy 2010). Similarly, despite his extensive experience within the jihadist movement, al-Zawahiri faced difficulties maintaining cohesion in al-Qaida after bin Laden’s death (Bacon and Arsenault 2019; Mendelsohn 2022). Leadership succession may thus trigger an authority crisis, a condition in which the new leader’s legitimacy must be constructed, defended, and continually reaffirmed.

We argue that, in the face of these vulnerabilities, successor leaders have strong incentives to visibly demonstrate their capacity to lead and signal their resolve. Internally, they must project strength to disloyal elites and skeptical commanders, reassuring the organization of their ability to maintain cohesion, discipline, and authority. Externally, successors are under pressure to project stability to both potential recruits and adversaries—such as state forces or rival rebel groups—who may interpret the leadership transition as a sign of internal weakness. In both spheres, the successor must counter perceptions that the organization is weakened (Price 2012; Abrahms and Potter 2015).

To overcome reputational deficits, one mechanism through which new militant leaders can signal resolve is by authorizing costly forms of violence, akin to how leadership transitions in states can provoke foreign policy changes or arms races (Rider 2013; Mattes, Leeds and Carroll 2015). In the case of armed groups, the use of gruesome violence can serve as an “unmistakable signal” of a leader’s commitment to the cause, precisely because of the costs it imposes (Weeraratne 2022, 7). Extra-lethal violence may thus send a powerful signal to both internal and external audiences. Internally, it can raise the perceived costs of dissent, deterring elite conspiracies while bolstering support among hard-line factions that favor extreme tactics. Externally, it can communicate strength and continuity, demonstrating to adversaries that the group remains capable of coordinated violence despite the recent change in leadership.

Hypothesis 1. Successor leaders are more likely than founding leaders to order beheadings.

rival factions.

Variation among Successors

Although all successor leaders may confront authority crises, not all successors are the same. Leaders are more than placeholders in an organizational timeline—they bring with them backgrounds, predispositions, and experiences that shape how they perceive threats and assess strategic options. Research on militant leaders shows that individual traits such as pre-war military experience, age, education, combat exposure, and foreign training significantly influence the strategies and tactics employed once in power (Doctor 2020; Huang, Silverman and Acosta 2022; Silverman, Acosta and Huang 2024). In the context of rebel succession, such individual-level characteristics help explain why some leaders resort to extreme violence—such as terrorism or suicide bombings—while others do not (Doctor, Hunter and Ligon 2024; Amjad et al. 2025). Variation in the use of extra-lethal violence, in other words, cannot be explained solely by successor status; it also depends on what the successor bring to the role.

We argue that a central factor shaping how successor leaders respond to authority crises is whether they previously led another armed organization before assuming their current position. This form of prior rebel leadership experience is distinct from general combat experience: it entails having exercised top-level authority—setting strategic direction, managing internal dynamics, and overseeing the use of violence. Experienced successors are not encountering the challenges of leadership for the first time; they bring with them a repertoire of leadership practices developed through direct experience in managing rebellion. Consequently, they are likely to interpret the threats and opportunities of succession through a different lens than those assuming organizational leadership for the first time.

First, successor leaders with prior rebel leadership experience may be more inclined to view violence against outsiders as a means of internal consolidation. Having previously governed another armed organization, such leaders are likely familiar with the internal tensions that threaten cohesion, such as factional rivalries or insubordination. In their earlier roles, they would have faced the challenge of maintaining authority under uncertainty, and some may have turned to violence as mechanisms for asserting control. This prior exposure can shape how they interpret the function of extra-lethal violence: not merely as a battlefield tactic, but as a political instrument to reinforce internal legitimacy. When confronted with skepticism or disloyalty following succession, these leaders should generally be more likely to draw on strategic lessons acquired through past leadership experience.

Similarly, experienced leaders are more likely to understand how brutality shapes external perceptions. Having previously led armed groups in confrontations with state forces or rival organizations, they may have witnessed firsthand how violence influences adversaries' behavior, or affects bargaining dynamics. Accordingly, they may perceive brutality not only as a tool of destruction,

but as a communicative act—one that projects strength and resolve. Such leaders may thus be more attuned to the strategic value of violence that shocks, intimidates, and reasserts the group’s stature in the eyes of external challengers.

At the same time, the fact that these leaders are no longer in command of their previous organizations suggests that their earlier tenures may have ended in failure—whether through internal ousting, arrest, or military defeat (Kreutz 2010; Prorok 2016). In this sense, prior leadership experience can be as much a marker of failure as a credential for learning. Certain forms of wartime authority may expose leaders to heightened scrutiny and sanction (Ortiz 2024), and such histories can exacerbate the successor’s authority crisis: they must not only contend with the usual burdens of replacing a founder, but also overcome reputational doubts linked to their personal track record (Lupton 2018; Renshon, Dafoe and Huth 2018). Their past, therefore, amplifies the reputational stakes of their successor status—they are tasked not only with consolidating authority in a new organization, but also with distancing themselves from the failures that marked their prior command.

This dual legacy of organizational learning and personal failure may make experienced successors particularly prone to the use of extra-lethal violence. As previous leaders, they understand how brutality can suppress dissent and deter adversaries. And they are acutely aware of the risks of losing power. As a result, they may not only see such violence as strategically advantageous but as essential to avoiding a repeat of past collapse. While inexperienced successors might regard extreme violence as counterproductive, or reputationally hazardous, experienced leaders are more likely to view it as a necessary instrument for navigating the fraught of succession.

Hypothesis 2. Successor leaders with prior rebel leadership experience are more likely than those without such experience to order beheadings.

Research Design

Our analysis draws on original data from the Jihadist Leaders Dataset (JLD) (Amjad et al. 2025), which provides novel information on top leaders of jihadist organizations active across multiple regions between 1976 and 2023. The JLD’s group sample, which includes militant Sunni and Shia Islamist organizations, is compiled from four major sources: the Armed Group Dataset (Malone 2022), the Mapping Militants Project (Crenshaw 2010), the Big, Allied, and Dangerous Database (Asal and Rethemeyer 2015), and the Jihadist Beheadings Dataset (Brzezinski 2024). The dataset identifies the top leader of each group as “the individual who exerts ultimate decision-making authority over major group policies” (Prorok 2016, 76). Drawing on sources in Arabic, French, German, Turkish, and Urdu, the JLD compiles detailed biographical information for each leader,

including their entry into and exit from power. All variables used in the analysis are derived from the JLD.

Dependent Variable

The primary outcome of interest is the use of beheadings by jihadist organizations during the tenure of individual leaders. Data on beheading incidents are drawn in part from data compiled by Brzezinski (2024), which covers 93 jihadist groups. Because this dataset does not include all groups in JLD, we supplemented it with additional data on jihadist beheading attacks (Berlin 2024). For JLD groups not covered in the original dataset, we reviewed organizational attacks recorded in the Global Terrorism Database (GTD) and the Armed Conflict Location and Event Data (ACLED) Project, and conducted extensive Boolean searches through thousands of news articles on NexisUni.

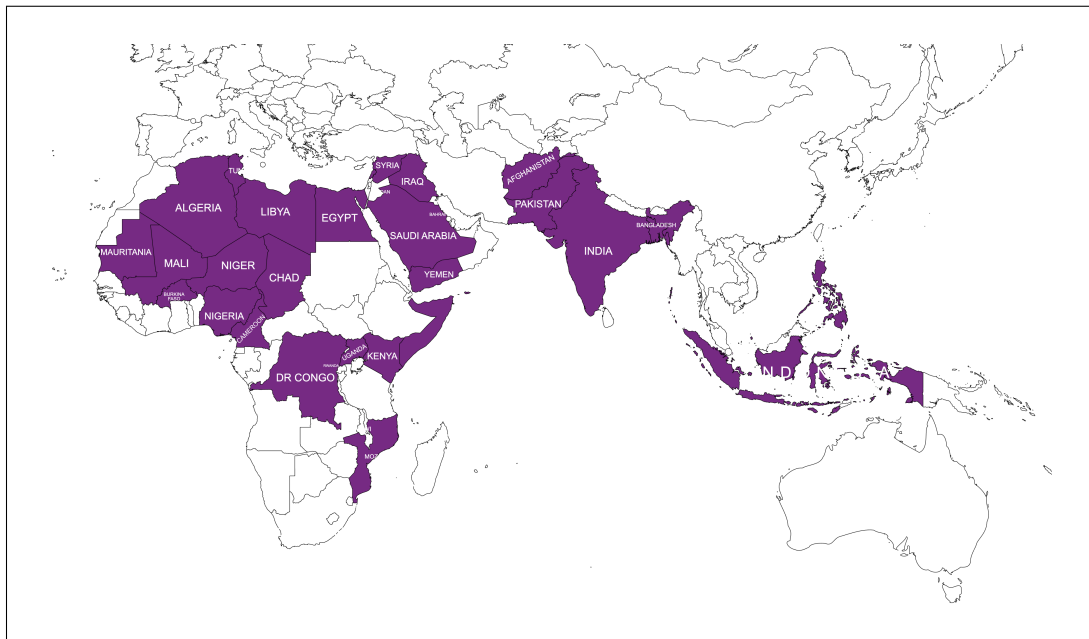


Figure 1. Countries with Jihadist Beheading Attacks, 2000-2019

The resulting analysis dataset includes 206 unique leaders across 108 organizations, spanning 1,305 leader-years. Among these, 42 organizations perpetrated at least one beheading. These incidents occurred under 78 distinct leaders, producing 214 leader-years—approximately 16 percent of the total—in which beheadings occurred. While the tactic is relatively rare, these 42 groups committed 1,726 documented beheadings, resulting in an estimated 4,380 (low) to 4,976 (high) casualties, underscoring the disproportionate human toll relative to the frequency of use.

Although the number of beheadings varies across leaders, most high-frequency incidents are

concentrated within five organizations: IS, the Taliban, al-Shabaab, Boko Haram, and Tehrik-i-Taliban Pakistan (TTP). Roughly half of the groups that employed beheadings did so only once or twice per leader-year. Accordingly, our main analysis focuses on the occurrence rather than the frequency of beheadings.

The dependent variable—*Beheading*—is a binary indicator coded 1 if a group perpetrated at least one beheading in a given leader-year, and 0 otherwise. This measure includes beheadings of both combatants and civilians. In robustness checks (Appendix 3-4-5), we replicate the main findings by disaggregating target types and using count models, including negative binomial regressions. The primary analysis employs logistic regression to estimate the effect of successor status and prior rebel leadership experience on the likelihood that a group used beheadings during a given leader-year.

Leader-Level Explanatory Variables

Our analysis includes two key leader-level explanatory variables: *successor leaders* and *prior rebel leadership experience*.

We define *successor leaders* as those who assumed command of a jihadist organization after its founding leader. This includes not only first successors but also subsequent leaders. The variable is coded as 1 if the leader was a successor and 0 if the founding leader. In the dataset, 43 percent of leaders are founders and 57 percent are successors; at the leader-year level, 51 percent of observations involve founders and 49 percent successors.

The second variable captures whether a leader had *prior rebel leadership experience* before assuming command of their current organization. Drawing on the JLD, we identify cases in which a leader previously held top-level command—defined as leading another non-state armed group—prior to their current role. For example, Iyad Ag Ghali, who later led Jama’at Nusrat al-Islam wal Muslimeen, had previously commanded Ansar al-Dine. The variable is coded as 1 if the leader previously served in a top leadership position in another rebel group, and 0 otherwise. This measure applies to both founders and successors. In our sample, 12 percent of leaders and 13 percent of leader-years involve individuals with such prior experience.

It is important to distinguish this measure from other forms of prior involvement in conflict. Previous leadership experience does not capture whether a leader previously engaged in combat or served in a state military. Instead, this measure specifically reflects prior experience exercising strategic command in an armed group, making it a measure of leadership, not general militancy.

Table 1 reports the proportion of leader-years involving beheadings across four types of jihadist leaders, categorized by successor status and prior leadership experience. The descriptive patterns provide initial support for our theoretical claims. Beheadings occurred in 36 percent of leader-

years under successors with prior leadership experience, compared to 21 percent under successors without such experience, and just 12–13 percent under founders. These patterns are consistent with both H1—that successor leaders are more likely than founders to employ extra-lethal violence—and H2—that prior leadership experience increases this likelihood among successors.

Examples from the data illustrate these dynamics. For instance, Harkat-ul-Mujahideen did not perpetrate beheadings under its founder Fazlur Rahman Khalil, but began doing so under his successor, Farooq Kashmiri. Similarly, Tehrik-i-Taliban Pakistan (TTP) refrained from using beheadings under Noor Wali Mehsud, a successor without prior leadership experience, but did so under Mullah Fazlullah, who brought such experience to the role.

Leader Type	No Beheading	At Least One Beheading
Founder with no leadership experience	87 percent <i>Harkat-ul-Mujahideen under Khalil</i> <i>Ansaru-Nigeria under Adam Kambar</i> <i>ETIM under Hasan Mahsum</i> <i>Indian Mujahideen under Reza Khan</i>	13 percent <i>Taliban under Mullah Omar</i> <i>AQIM under Hassan Hattab</i> <i>Boko Haram under Mohammed Yusuf</i> <i>Lashkar-e-Taiba under Hafiz Saeed</i>
Founder with leadership experience	88 percent <i>Ras Kamboni Brigades under al-Turki</i> <i>MILF under Hashim Salamat</i> <i>Jundallah-Pakistan under Hakimullah</i> <i>Hizbul Islam under Abu Bakar</i>	12 percent <i>AQAP under Nasir al-Wuhayshi</i> <i>ADF under Jamil Mukulu</i> <i>Caucasus Emirate under Umarov</i> <i>Katiba Muhajireen under al-Shishani</i>
Successor with no leadership experience	79 percent <i>Haqqani Network under Sirajuddin</i> <i>TTP under Noor Wali Mehsud</i> <i>Jemaah Islamiya under Ba'asyir</i> <i>Badr Brigade under al-Amiri</i>	21 percent <i>Abu Sayyaf under Khadaffy Janjalani</i> <i>al-Shabaab under Ahmad Umar</i> <i>Lashkar-e-Islam under Mangal Bagh</i> <i>Harkat-ul-Mujahideen under Kashmiri</i>
Successor with leadership experience	64 percent <i>al-Qaida under Ayman al-Zawahiri</i> <i>Ansar al-Islam under al-Shafi</i> <i>al-Mourabitoun under Belmokhtar</i>	36 percent <i>TTP under Mullah Fazlullah</i> <i>Islamic State under al-Baghdadi</i> <i>Hayat Tahrir al-Sham under al-Jawlani</i>

Table 1. Beheadings, by Leadership Type and Prior Rebel Leadership Experience

Note: Table presents the proportion of leader-years in which jihadist organizations perpetrated no beheadings or at least one beheading.

Leader-Level Controls

We include a set of leader-level control variables to address potential confounders that may influence both our key independent variables—successor status and prior rebel leadership experience—and the likelihood that a group engages in beheadings.

First, we control for *leadership tenure*, measured as the number of years a leader has held power. This variable increases incrementally each year and is logged in the models to account

for non-linear effects. Leaders with longer tenures might be more likely to oversee violent acts, including beheadings, simply due to their extended time in office. Tenure may also correlate with both successor status and prior leadership experience—e.g., experienced leaders may survive longer in power.

Second, we include the *age* at which a leader assumed power, also logged. Age is often linked to leadership style, risk tolerance, and ideological rigidity (Silverman, Acosta and Huang 2024). Younger leaders may pursue more aggressive tactics, while older leaders may be more cautious. If age systematically differs between founders and successors—for example, if successors are older than founders—it could confound the relationship between leadership background and tactical decisions.

Third, we control for whether the leader spent *time in prison* prior to assuming command. Prior incarceration may function as a site for ideological radicalization or for building militant credentials and networks (Mironova 2019). It may also signal a deeper commitment to militancy, increasing the likelihood of adopting extreme tactics. At the same time, prison experience may correlate with advancement through the ranks, and thus with being a successor or having prior leadership experience.

Finally, we include binary indicators for whether the leader *pledged allegiance* to al-Qaida or IS. Such pledges may reflect stronger ideological ties to global jihadist movements and greater adherence to tactical norms involving excessive violence since both AQ and IS are associated with the routinized use of beheadings. Moreover, leaders with prior rebel leadership experience may be more likely to pledge to AQ or IS due to preexisting networks or affiliations developed in earlier roles.

Group- and State-Level Controls

We also include control variables at the group- and state- levels. At the group level, we control for whether the organization was designated as a Foreign Terrorist Organization (*FTO*) by the United States. Groups employing extreme forms of violence may be more likely to receive this designation due to the brutality of their tactics. At the same time, groups led by certain types of leaders—particularly those with prior rebel leadership experience—may attract FTO status due to the leader’s prior notoriety. FTO designation, therefore, may correlate with both leadership characteristics and tactical behavior.

Second, we control for the group’s *ideology*, coded as a categorical variable distinguishing among Salafi-Jihadist, Salafi-Nationalist, and Other. Salafi-Jihadist groups follow Salafist doctrine and pursue transnational objectives, while Salafi-Nationalist groups pursue more localized goals (Brzezinski 2024). The “Other” category includes jihadist organizations with alternative ideolog-

ical orientations, such as Shia or Deobandi groups. Ideology can shape both the type of leaders promoted and the tactics deemed acceptable. For instance, Salafi-Jihadist groups may be more inclined to select leaders with prior experience and established links to transnational networks like IS or AQ, which may also increase the likelihood of endorsing extreme violence.

All leader- and group-level control variables described here are drawn from the JLD. Descriptive statistics are presented in Appendix 1. In supplemental models reported in Appendix 2, we include additional controls to test the robustness of our findings. These include indicators for whether the group exercised *territorial control*, the estimated *size* of the group, the number of *competing armed groups* active in the same country, and whether the country experienced a *regime change* in the past five years. These variables are sourced from the dataset compiled by Brzezinski (2024).

Results

Successors Are More Likely to Behead

We begin by evaluating Hypothesis 1, which posits that successor leaders are more likely than founders to oversee the use of beheadings. Table 2 presents results from a series of logistic regression models predicting whether a jihadist group committed at least one beheading in a given leader-year.

Across all three models, the coefficient on the *Successor leader* variable is positive, statistically significant, and substantively meaningful. In Model 1, which includes only the main independent and dependent variables, being a successor is associated with a significant increase in the likelihood of beheading ($p < 0.01$). This effect remains robust in Model 2, which adds the full set of leader-level controls ($p < 0.01$), and persists—albeit with slightly reduced magnitude—in Model 3, which incorporates both leader- and group-level controls ($p < 0.01$). These results offer strong empirical support for H1. Even after accounting for tenure, age, ideological orientation, FTO designation, and pledges to AQ or IS, successor leaders remain significantly more likely to preside over beheadings.

Substantively, holding all other variables constant, the predicted probability that a group perpetrates a beheading in a given leader-year increases from approximately 8.9% under a founding leader to 16.2% under a successor. This almost twofold increase highlights the substantive importance of leadership transitions: successor leaders are not only statistically more likely to oversee beheadings, but the magnitude of the effect suggests meaningful shifts in the tactical behavior of jihadist organizations depending on who holds power.

Table 2. Logit Models of Jihadist Beheadings (H1)

	Dependent Variable: Beheading (Dummy)		
	Model 1	Model 2	Model 3
Successor leader	0.662*** (0.154)	0.676*** (0.176)	0.497*** (0.192)
Previous rebel leader experience		0.450* (0.236)	0.118 (0.248)
Leadership tenure		−0.037 (0.102)	−0.112 (0.112)
Age		−1.036*** (0.368)	−0.811** (0.391)
Prison time before leadership		−0.030 (0.202)	0.068 (0.218)
Leader pledged to AQ		1.507*** (0.233)	0.974*** (0.250)
Leader pledged to IS		1.583*** (0.310)	1.169*** (0.322)
Salafi-jihadist			0.636*** (0.229)
Salafi-nationalist			−0.694 (0.623)
FTO designated			0.904*** (0.198)
Constant	−1.900*** (0.118)	1.688 (1.307)	0.318 (1.433)
Observations	1,221	1,066	1,066
Log Likelihood	−557.247	−453.023	−431.044
Akaike Inf. Crit.	1,118.493	922.046	884.089

Note 1: ***p < 0.01, **p < 0.05, *p < 0.1

Note 2: The reference category for group ideology is “Other” ideology.

Successors With Prior Leadership Experience Are Even More Likely to Behead

To evaluate Hypothesis 2, we estimate a series of interaction models assessing whether successor leaders with prior rebel leadership experience are more likely to oversee beheadings than other types of leaders. Table 3 presents results from three logistic regression models that include an interaction term between *Successor leader* and *Previous rebel leader experience*.

In all three models, the interaction term is positive, statistically significant, and substantively large. In Model 4, the interaction coefficient is 0.938 ($p < 0.05$), increasing to 1.186 in Model 5 and 1.053 in Model 6 (both $p < 0.05$), even after accounting for leader- and group-level controls. These findings indicate that the effect of being a successor leader is conditional on prior rebel leadership experience: successor leaders with such experience are significantly more likely to preside over beheadings than founders or inexperienced successors. The results strongly support H2 and are consistent with our theoretical claim that prior leadership roles shape how successors interpret and respond to the authority challenges they inherit.

Figure 2 visualizes the interaction by plotting the predicted probability of beheading across combinations of successor status and prior leadership experience. The conditional effect is clear: predicted probabilities are relatively low and similar among founders, regardless of experience, but diverge sharply among successors. Successors without prior leadership experience have an estimated probability of beheading just under 15 percent, while this figure rises to approximately 32 percent for those with prior rebel leadership experience—more than doubling the likelihood of a beheading.

Robustness

Our results are robust to alternative modeling approaches, including negative binomial models (Appendix 3) and disaggregating victim types—combatants and civilians (Appendices 4–5).

We also estimate additional models incorporating a broader set of group- and state-level controls from Brzezinski (2024). When re-estimating our models with these covariates, the core results remain virtually unchanged (Appendix 2). The effects of successor status and its interaction with prior rebel leadership experience are both statistically and substantively consistent across specifications. Notably, our models also replicate Brzezinski's findings regarding the positive association between territorial control, group size, and the likelihood of beheadings. While the inclusion of these additional controls reduces the sample size—due to more limited coverage in Brzezinski's dataset compared to the JLD—the key relationships persist. This consistency reinforces the conclusion that our findings are not artifacts of sample composition or model specification.

Table 3. Logit Models of Jihadist Beheadings with Interaction (H2)

	Dependent Variable: Beheading (Dummy)		
	Model 4	Model 5	Model 6
Successor leader	0.535*** (0.167)	0.480** (0.190)	0.336 (0.204)
Previous rebel leader experience	−0.155 (0.345)	−0.205 (0.372)	−0.477 (0.384)
Successor * Previous leader experience	0.938** (0.443)	1.186** (0.471)	1.053** (0.479)
Leadership tenure		−0.022 (0.104)	−0.094 (0.114)
Age		−1.086*** (0.371)	−0.828** (0.392)
Prison time before leadership		−0.006 (0.203)	0.094 (0.218)
Leader pledged to AQ		1.545*** (0.235)	1.011*** (0.252)
Leader pledged to IS		1.632*** (0.313)	1.214*** (0.324)
Salafi-jihadist			0.660*** (0.229)
Salafi-nationalist			−0.727 (0.623)
FTO designated			0.861*** (0.201)
Constant	−1.878*** (0.127)	1.949 (1.317)	0.439 (1.433)
Observations	1,221	1,066	1,066
Log Likelihood	−553.408	−449.703	−428.531
Akaike Inf. Crit.	1,114.816	917.407	881.062

Note 1: ***p < 0.01, **p < 0.05, *p < 0.1

Note 2: The reference category for group ideology is “Other” ideology.

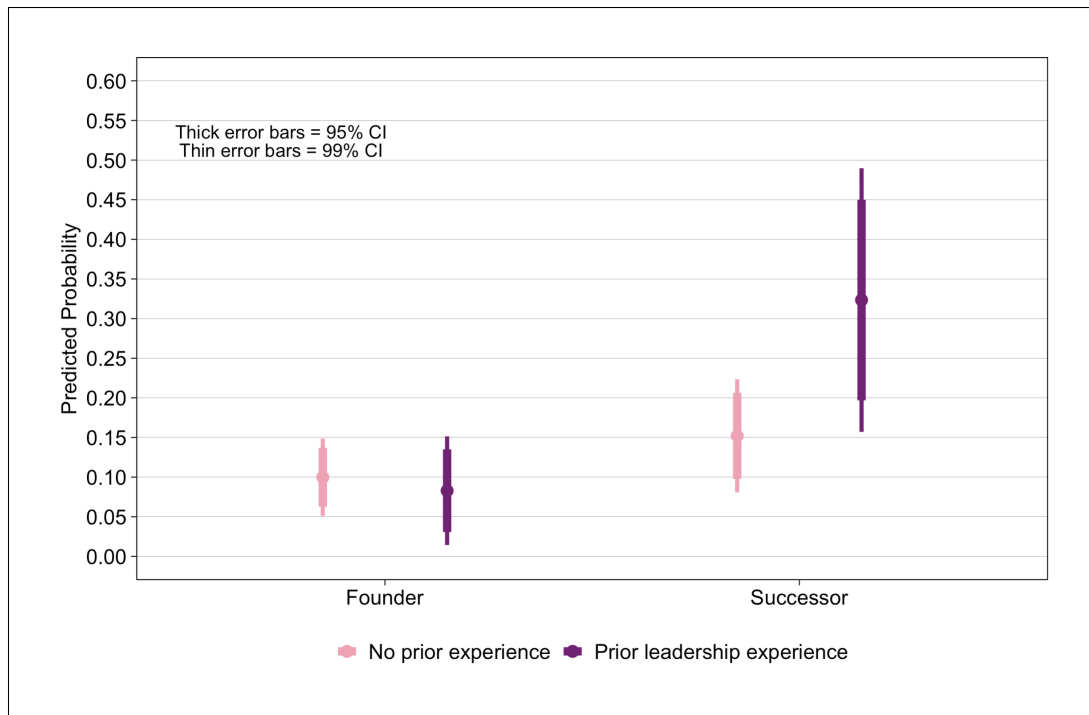


Figure 2. Effects of Successor Status and Prior Leadership Experience on Beheadings
Note: Results are predicted probabilities from Model 6 in Table 3. Continuous control variables are kept at their mean. Binary control variables are kept at their median. Group ideology is kept at “Salafi-Jihadist”.

Substantiating Causal Mechanisms

Evidence of Leader Sanctioning of Beheadings

Our theory posits that successor leaders authorize extreme forms of violence to consolidate authority and signal resolve to external audiences. A potential alternative explanation for the patterns we observe is that beheadings were not sanctioned by leadership but instead carried out by rogue fighters acting independently of the command structure. Although it is difficult to verify whether leaders personally ordered these acts, public claims of responsibility by the organization offer a useful proxy. If groups openly take credit for beheadings, this undermines the notion that such violence occurred without leadership approval.

Drawing on data from Brzezinski (2024) regarding organizational claims of beheadings, we find that successor leaders, especially those with prior leadership experience, are significantly more likely to claim responsibility for these acts than other leaders. This pattern supports our argument that successors strategically employ extra-lethal violence to project strength and reaffirm control, rather than merely tolerating decentralized brutality.

Why Are Successors More Likely to Behead? (H1)

Our theory posits that successor leaders are more likely than founders to authorize extra-lethal violence because they face authority crises following leadership transitions. However, this relationship could also be consistent with alternative explanations. One possibility is ideological radicalization over time: successor leaders may appear more violent because the organizations they inherit have become more extreme. A second is strategic escalation, where successors take power during more intense phases of conflict. A third involves selection effects: more violent organizations may be both more prone to leadership change and more likely to appoint aggressive successors.

To adjudicate among these explanations, we first examine whether the nature of the leadership transition moderates the effect of successor status. Specifically, we compare successor leaders who come to power after the sudden removal of a predecessor—via killing or arrest—to those who assume power following more orderly exits, such as death by natural causes or voluntary departure. This variation captures differences in the severity of the authority crisis. If beheadings are indeed used to compensate for authority deficits, we should observe stronger effects following abrupt transitions. Indeed, we find that successors who inherit power after sudden removals are significantly more likely to oversee beheadings than those following more stable transitions (Appendix 6). This pattern supports our theoretical mechanism and challenges rival explanations. Ideological radicalization or conflict escalation should not hinge on how leadership transitions occur, and selection effects cannot easily explain variation in succession type among successors.

Our second test evaluates whether the successor effect diminishes over time, as would be expected if beheadings are a response to short-term authority crises. We interact the successor indicator with leadership tenure, hypothesizing that successors may rely on beheadings early in their tenure but reduce their use as they consolidate control. The results support this claim: the probability of beheading is highest at the start of a successor's tenure and declines steadily thereafter (Appendix 7). This temporal dynamic is difficult to reconcile with ideological or structural explanations, which would predict stable or increasing use of violence over time. Instead, it reinforces our argument that successor leaders turn to extra-lethal violence as a strategic response to the instability that accompanies leadership transition.

Why Are Experienced Successors Even More Likely to Behead? (H2)

Our second hypothesis posits that successor leaders with prior rebel leadership experience are more likely to employ extra-lethal violence because they are uniquely positioned and incentivized to recognize and deploy such tactics as strategic tools for consolidating authority and signaling resolve. This mechanism highlights the formative role of previous leadership in shaping how successors interpret and respond to the authority crisis. However, the observed interaction between

successor status and prior leadership experience could also reflect alternative explanations. These leaders may be more ideologically extreme (e.g., through religious training), operationally more skilled in violence (e.g., through military backgrounds), or more deeply embedded in transnational jihadist networks (e.g., via foreign fighter experience), all of which could influence both their selection and their tactical preferences.

To adjudicate between these competing explanations, we conduct a series of interaction analyses examining whether other leadership traits—religious education, formal military experience, and foreign fighter history—moderate the relationship between successor status and beheadings in a manner similar to prior rebel leadership experience.

First, we test whether religious education—used here as a proxy for doctrinal training and potential ideological rigidity—moderates the successor-beheading relationship. If prior leadership experience simply reflects stronger ideological commitment to extreme violence, religiously trained successors should show similar patterns. Yet our analysis reveals no such effect: religious education does not increase the likelihood that a successor oversees beheadings (Appendix 8). This suggests that the influence of leadership experience is not reducible to ideological extremism, but instead reflects the strategic learning associated with prior command.

Second, we assess whether formal military experience—often used to proxy tactical skill and organizational competence—functions similarly. If prior rebel leadership is merely capturing operational capability, then successors with military training should behave comparably. However, interaction models show no significant effect: successor leaders with military backgrounds are no more likely to preside over beheadings than those without such training (Appendix 9). This undermines the argument that the observed effect is driven by technical proficiency in violence.

Finally, we examine whether foreign fighter experience—indicating exposure to transnational jihadist networks like al-Qaida or IS—produces similar patterns. If leadership experience simply tracks embeddedness in global jihadi communities, we would expect similar effects among foreign fighters. Instead, we find the opposite: successors with foreign fighter backgrounds are less likely to be associated with beheadings (Appendix 10). This not only weakens the network exposure explanation but reinforces our claim that it is the experience of commanding a rebel organization—not merely participating in armed conflict—that shapes a successor’s propensity to use extreme violence as a tool of internal consolidation.

Conclusion

Extra-lethal violence constitutes a key feature of contemporary conflicts. Focusing on beheadings, this paper analyzed how leader-level variation is associated with the use of extra-lethal violence by jihadist organizations. Specifically, we theorized that successor leaders are more likely to engage

in extreme violence to signal resolve and increase their authority. Drawing on novel jihadist leader data, the findings underscore the importance of previous leadership experience, showcasing the role that background experiences may play in affecting armed group leaders' preferences when they take power.

We assessed our theory with a sample of jihadist groups, among the most likely type of armed group to use beheading. However, our argument that successor leaders face acute pressure to shore-up support, which leads them to use extra-lethal violence, should generalize to armed groups more broadly. Successors in non-jihadist groups may use other forms of extra-lethal violence, such as immolation or crucifixion, but the tool remains available for successors to assert their authority. This variation can be studied as more and better data about various forms of extra-lethal violence are published. In doing so, scholars can assess the broader applicability of our argument and contribute to emerging work on why militant actors engage in spectacular violence despite its costs (Badillo-Sarmiento and Trejos-Rosero 2025).

These findings open several promising avenues for future research. One concerns organizational cohesion: armed groups frequently undergo fragmentation and splintering, and future work could examine whether such breakdowns are more likely under successor leaders (Doctor 2020). These same authority deficits may also undermine successors' ability to credibly negotiate peace agreements until consolidating their authority. Another line of inquiry could investigate the timing and target selection of extra-lethal violence, as well as how such tactics shape civilian perceptions. Understanding when, why, and against whom these forms of violence are used can offer deeper insight into the strategic logic of spectacular brutality.

Finally, future research can investigate how armed groups respond to state pressure. Leadership decapitation is a prominent state counterterrorism tool (Price 2012; Jordan 2014). Our findings highlight how leadership decapitation strikes can lead to increases in extreme forms of violence. Decapitation strikes also have implications for the way in which armed groups engage with civilian populations (Bauer, Reese and Ruby 2022). In multiple cases, jihadist organizations often beheaded local civilians to deter spying. Further analysis can study whether leadership decapitation has the unintended effect of increasing beheadings, which complicates its use as a counterterrorism tactic.

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